Student Reflection Notes: Cloud Computing Mini project

Course: CS3081 (Cloud Computing)

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Github → github/govindaaaaa/cloud-computing-assignment.

In this project, my goal was to build a complete cloud system that takes normal voice commands ("wake me up at 7 am or" what is the weather today?") and turns them into clean Json data. also I have learned how seq. level knowledge works - training a small student (pythia-70m) and on teacher's (phi-3.5) generated text, not it's weights.

& also learnt object. storage (seaweedFS/S3), running kubernetes with k3d (problems, resources, limits).

challanges and Solutions →

→ Teacher added prose after it's Json (17% valid) → used olama format: "json" 100% valid.

→ Student scored 0% → a trailing space broke tokenization; ending it as "Json". fixed it (0 → 95%)

→ My laptop could not ~~run~~ run docker (no WSL2/ virtualization) → ran the k3d cluster on GitHub actions.

→ kept the image ≤ 2GB with CPU-only pytorch, weights as Release assets, not in git.

Findings →

students full frame 43% vs teachers 48%, 99% valid

Jsod - a 70m model underfits cluster: image 1.74GB, cold start 8s, pas7805 at 2.41/ps load balance 51/49, 510 99.1%.

- 0.91 IdR per 1000 requests. dacl 6/6.

Improvements →

requests batching, more epochs/a larger students, a shared work queue, a multi-node cluster with a Pad disruptionBudget, and per-service scoped S3 keys.